

COMP4035

Networking Coursework

Section-A	Connect a Wired and Wireless LAN	10 marks
Section-B	Subnetting scheme	15 marks

Learning Outcome:

Understand and apply simple data transmission, network layers, Internet protocol, and routing and switching

Deadline: Friday Week-12

Submission Instructions:

Upload a copy of both the *simulation file* and the *Word document* on Moodle. Your word file should have the answers for part A and the tables and report for part B

Note:

You will need the associated packet tracer file to do section-A of the coursework

Section-A**Connect a Wired and Wireless LAN****Addressing Table**

Device	Interface	IP Address	Connects To
Cloud	Eth6	N/A	F0/0
	Coax7	N/A	Port0
Cable Modem	Port0	N/A	Coax7
	Port1	N/A	Internet
Router0	Console	N/A	RS232
	F0/0	192.168.2.1/24	Eth6
	F0/1	10.0.0.1/24	F0
	Ser0/0/0	172.31.0.1/24	Ser0/0
Router1	Ser0/0	172.31.0.2/24	Ser0/0/0
	F1/0	172.16.0.1/24	F0/1
WirelessRouter	Internet	192.168.2.2/24	Port 1
	Eth1	192.168.1.1	F0
Family PC	F0	192.168.1.102	Eth1
Switch	F0/1	172.16.0.2	F1/0
Netacad.pka	F0	10.0.0.254	F0/1
Configuration Terminal	RS232	N/A	Console

Objectives**Part 1: Connect to the Cloud****Part 2: Connect Router0****Part 3: Connect Remaining Devices****Part 4: Verify Connections****Part 5: Examine the Physical Topology****Background**

When working in Packet Tracer (a lab environment or a corporate setting), you should know how to select the appropriate cable and how to properly connect devices. This activity will examine device configurations in Packet Tracer, selecting the proper cable based on the configuration, and connecting the devices. This activity will also explore the physical view of the network in Packet Tracer.

Instructions

Part 1: Connect to the Cloud

Step 1: Connect the cloud to Router0.

- a. At the bottom left, click the orange lightning icon to open the available **Connections**.
- b. Choose the correct cable to connect **Router0 F0/0** to **Cloud Eth6**. **Cloud** is a type of switch, so use a **Copper Straight-Through** connection. If you attached the correct cable, the link lights on the cable turn green.

Step 2: Connect the cloud to Cable Modem.

Choose the correct cable to connect **Cloud Coax7** to **Modem Port0**.

If you attached the correct cable, the link lights on the cable turn green.

Part 2: Connect Router0

Step 1: Connect Router0 to Router1.

Choose the correct cable to connect **Router0 Ser0/0/0** to **Router1 Ser0/0**. Use one of the available **Serial** cables.

If you attached the correct cable, the link lights on the cable turn green.

Step 2: Connect Router0 to netacad.pka.

Choose the correct cable to connect **Router0 F0/1** to **netacad.pka F0**. Routers and computers traditionally use the same wires to transmit (1 and 2) and receive (3 and 6). The correct cable to choose consists of these crossed wires. Although many NICs can now autosense which pair is used to transmit and receive, **Router0** and **netacad.pka** do not have autosensing NICs.

If you attached the correct cable, the link lights on the cable turn green.

Step 3: Connect Router0 to the Configuration Terminal.

Choose the correct cable to connect **Router0 Console** to **Configuration Terminal RS232**. This cable does not provide network access to **Configuration Terminal**, but allows you to configure **Router0** through its terminal.

If you attached the correct cable, the link lights on the cable turn black.

Part 3: Connect Remaining Devices

Step 1: Connect Router1 to Switch.

Choose the correct cable to connect **Router1 F1/0** to **Switch F0/1**.

If you attached the correct cable, the link lights on the cable turn green. Allow a few seconds for the light to transition from amber to green.

Step 2: Connect Cable Modem to Wireless Router.

Choose the correct cable to connect **Cable Modem Port1** to **Wireless Router Internet** port.

If you attached the correct cable, the link lights on the cable will turn green.

Step 3: Connect Wireless Router to Family PC.

Choose the correct cable to connect **Wireless Router Ethernet 1** to **Family PC**.

If you attached the correct cable, the link lights on the cable turn green.

Part 4: Verify Connections

Step 1: Test the connection from Family PC to netacad.pka.

- a. Open the **Family PC** command prompt and ping **netacad.pka**.
- b. Open the **Web Browser** and the web address **http://netacad.pka**.

Step 2: Ping the Switch from Home PC.

Open the **Home PC** command prompt and ping the **Switch** IP address of to verify the connection.

Step 3: Open Router0 from Configuration Terminal.

- a. Open the **Terminal** of **Configuration Terminal** and accept the default settings.
- b. Press **Enter** to view the **Router0** command prompt.
- c. Type **show ip interface brief** to view interface statuses.

Part 5: Examine the Physical Topology

Step 1: Examine the Cloud.

- a. Click the **Physical Workspace** tab or press **Shift+P** and **Shift+L** to toggle between the logical and physical workspaces.
- b. Click the **Home City** icon.
- c. Click the **Cloud** icon.

How many wires are connected to the switch in the blue rack?

- d. Click **Back** to return to **Home City**.

Step 2: Examine the Primary Network.

- a. Click the **Primary Network** icon. Hold the mouse pointer over the various cables.

What is located on the table to the right of the blue rack?

- b. Click **Back** to return to **Home City**.

Step 3: Examine the Secondary Network.

- a. Click the **Secondary Network** icon. Hold the mouse pointer over the various cables.

Why are there two orange cables connected to each device?

- b. Click **Back** to return to **Home City**.

Step 4: Examine the Home Network.

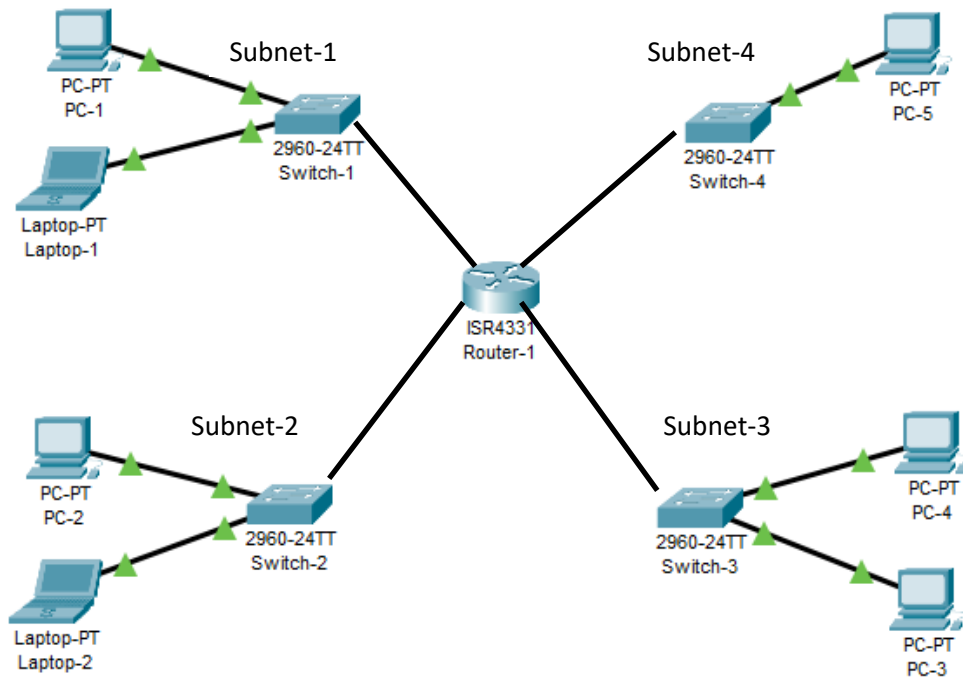
- a. Click the **Home Network** icon.

Why is there no rack to hold the equipment?

- b. Click the **Logical Workspace** tab to return to the logical topology.

Section-B

Subnetting scheme



The above network consists of 4 subnets interconnected by a router (Router-1). Each of the subnets have its own switch (e.g. Switch-1) and a number of PCs and laptops connected via the switch. Your task is to assign an addressing scheme for the above network. You have been assigned a Class B address with the first two numbers of the dotted decimal prefix is based on the numeric equivalent of the first two letters of your surname such that A=130, B=131, C=132 and so on.

Task: Design an appropriate subnetting scheme for the above network justifying the number of bits allocated for the network and host ids. Complete the tables below with the addressing information. Finally, write a short report (maximum 1 page) explaining the steps.

Class B Address	
Number of host bits	
Subnet Mask	

Addressing Table

Device	IP Address
Subnet-1 Network Address	
Switch-1 VLAN1	
PC-1	
Laptop-1	
Subnet-2 Network Address	
Switch-2 VLAN1	
PC-2	
Laptop-2	
Subnet-3 Network Address	
Switch-3 VLAN1	
PC-3	
PC-4	
Subnet-4 Network Address	
Switch-4 VLAN1	
PC-5	

Note: This is an individual assignment. You should produce your own solution and not share your work with other students.

Any evidence of copying will be forwarded to the Academic Conduct Officer.

You are expected to have familiarised yourself with the University's Academic Conduct regulations.

<https://www.brookes.ac.uk/students/sirt/student-conduct/>
